Samuel Indyk

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EDUCATION

University of Austin — B.S. Liberal Studies, Concentration in Applied Engineering | *Austin, TX | Sep 2024 – Jun 2028*
Awards: Founders Scholarship; Lonsdale-Magaro Scholarship
Coursework: Analysis of Circuits, Software Engineering, The Rise & Fall of Ancient Rome, The Sublime & The Beautiful

EXPERIENCE

Software Engineering Intern — *The Boring Company | Bastrop, TX | Jun 2026 – Present*
- Created Boreview, an interactive web-based 3D CAD platform that links I/O points to underlying hardware for TBC's machines, incorporating custom mesh-rendering server infrastructure and automated SolidWorks PDM sync.
- Built continuous network monitoring (MQTT, InfluxDB, Grafana) for tunnel boring machine camera and switch networks, with historical uptime and per-port diagnostics.
- Architected and deployed an enterprise-wide factory display system featuring an interactive map interface and self-healing kiosk agent.
- Migrated machine authentication to Django/PostgreSQL with training-gated access and role-based permissions.
- Assisted in developing depth-camera gap sensing systems by correcting rotation mathematics, engineering a self-cleaning mechanism, and conducting daily manual tunnel measurements to validate algorithm accuracy.

Policy Intern — *White House Office of Science and Technology Policy | Washington, DC | Jun 2025 – Aug 2025*
- Contributed to executive order development by gathering data and supporting policymakers.
- Drafted and coordinated public government communications and resource offerings.
- Supported the OSTP Director with scheduling, travel, and meetings across federal and non-USG stakeholders; hosted White House visitors, including foreign dignitaries.

Research Intern — *AlphaX | Austin, TX | Nov 2024 – Feb 2025*
- Designed robotics curricula for gt.school and Alpha High School based on research into best practices.

Defense Team Intern — *8VC | Austin, TX | Dec 2024 – Jan 2025*
- Analyzed and wrote on government contracting business models and authored a quarterly report distributed to LPs

EXTRACURRICULARS

Austin Union Society — *President | austinunion.org | Jun 2025 – Present*
- Conducting "Open Forum" events monthly and hosting guests such as Jonathan Pageau and Phil Magness.

UATX-Cavators — Mechanical and Software Lead | *Jun 2026 – Present*
- Primary slurry micro-TBM designer for The Boring Company's annual Not-A-Boring-Competition

PUBLICATIONS

S. Gaalema, **S. Indyk**, and C. Staley, "Reduced-Mass Orbital AI Inference via Integrated Solar, Compute, and Radiator Panels," arXiv:2604.07760 [cs.DC], Apr 2026.
- Co-developed the Integrated Solar Compute Radiator (ISCR) architecture, which fuses solar cells, vapor-chamber radiators, and compute ICs into a single panel to reach ~500 W/kg specific power — roughly 5× conventional designs.
- Modeled orbital thermal balance, power budget, and LLM inference throughput for a 16 MW, 16,000-panel satellite.

SKILLS

Programming & Data: Python, TypeScript, R, Django, TanStack, Docker, Structured Text
Engineering Tools: SolidWorks, Fusion 360, ANSYS, TwinCAT
Technical: Lathe and CNC operation, manual manufacturing and fabrication, oscilloscope / RF, soldering